ISO 32000-2 · PDF 2.0

Everything a page can do.

A suite of twenty-five example documents, each built from raw operators.

THE COLLECTION

02 Standard 14 fonts

03 Typography

04 Text render modes

05 Vector graphics

06 Patterns & shadings

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08 Colour spaces

09 Images

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20 Report

21 Brochure

22 Invoice

23 Résumé

24 Poster

25 Dashboard

26 Repacked

Each tile links to its sample file (GoToR). Open this document next to the other PDFs in the samples folder.

generated by tools/AdoBye.Samples · repeatable · no stock images

The standard 14 fonts

Every conforming reader supplies these Type 1 fonts; no embedding needed.

01 Helvetica
Sphinx of black quartz, judge my vow 0123

02 Helvetica-Bold
Sphinx of black quartz, judge my vow 0123

03 Helvetica-Oblique
Sphinx of black quartz, judge my vow 0123

04 Helvetica-BoldOblique
Sphinx of black quartz, judge my vow 0123

05 Times-Roman
Sphinx of black quartz, judge my vow 0123

06 Times-Bold
Sphinx of black quartz, judge my vow 0123

07 Times-Italic
Sphinx of black quartz, judge my vow 0123

08 Times-BoldItalic
Sphinx of black quartz, judge my vow 0123

09 Courier
Sphinx of black quartz, judge my vow 0123

10 Courier-Bold
Sphinx of black quartz, judge my vow 0123

11 Courier-Oblique
Sphinx of black quartz, judge my vow 0123

12 Courier-BoldOblique
Sphinx of black quartz, judge my vow 0123

13 Symbol
αβχδεφγηιφ ΑΒΧΔΕΦΓΗΙϑ κλμνοπ

14 ZapfDingbats
❁✲❃❉❋❄✳✺✻✼✽●❍■❑❒❑❒▲▼◆❖◗▮▮ 🖉🖋🖌✓✔✕✖✗✘✢

TEXT STATE OPERATORS ON TIMES AND HELVETICA

Word spacing adds space between words
Horizontal scaling 60% then 120%
Character spacing
Rise: base$^{super}_{sub}$ and back.
Italic via a second Tf, same T leading.*

Embedded typography

Source Serif 4, Source Sans 3 and Source Code Pro: embedded OpenType (CFF) subsets, GPOS kerning.

PAIR KERNING, OFF VERSUS ON

AVATAR To

Kerning off

AVATAR To

Kerning on (GPOS kern feature)

ALIGNMENT

LEFT

Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath the surface. A well-set page offers a clear entry point

CENTER

Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath the surface. A well-set page offers a clear entry point

RIGHT

Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath the surface. A well-set page offers a clear entry point

JUSTIFY

Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath the surface. A well-set page offers a clear entry point

COLUMNS AND A DROP CAP

Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath the surface. A well-set page offers a clear entry point, a steady measure, and enough air around each block for the eye to rest before it moves on. Kerning adjusts the space between particular pairs of glyphs, so that the optical colour of a line stays even whether it contains a wide W or a narrow l. Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath the surface. A well-set page offers a clear entry point, a steady measure, and enough air around each block for the eye to rest before it moves on. Kerning adjusts the space between particular pairs of glyphs, so that the optical colour of a line stays even whether it contains a wide W or a narrow l. Good typography is invisible. The reader notices the argument, never the letterforms; spacing, rhythm and contrast do their work quietly beneath

TJ ARRAY KERNING

```
BT
  /F1 12 Tf
  1 0 0 1 72 720 Tm
  [<0041> -80 <0056> 40 <0041>] TJ
ET
```

Scale, tracking, glyph coverage

Hamburgefonstiv 0123456 46 pt

Hamburgefonstiv 0123456789 36 pt

Hamburgefonstiv 0123456789 24 pt

Hamburgefonstiv 0123456789 16 pt

Hamburgefonstiv 0123456789 12 pt

Hamburgefonstiv 0123456789 9 pt

LETTERSPACING (TC)

CAPITALS NEED AIR Tc = 0

CAPITALS NEED AIR Tc = 1.5

CAPITALS NEED AIR Tc = 4

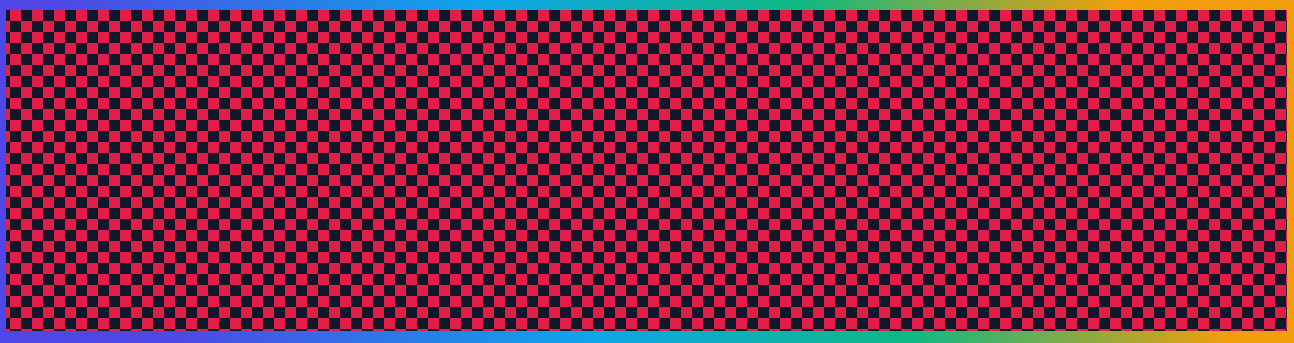
GLYPH COVERAGE

Latin: Zażółć gęślą jaźń — café, naïve, Œuvre, ß, ñ, å, ø, Þ

Greek: Αλφάβητος · α β γ δ ε ζ η θ · Σ Ω Δ

Cyrillic: Съешь же ещё этих мягких французских булок

Symbols: © ® ™ § ¶ † ‡ • … « » ‹ › € £ ¥ ¢ ° ± × ÷ ≈ ≠ ≤ ≥ ∞ → ←

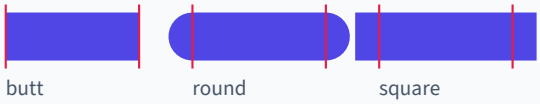


Every glyph is positioned and rotated individually along the arc

Vector graphics

Everything on this page is drawn with path operators: lines, Bézier curves, strokes and fills.

LINE CAPS



LINE JOINS

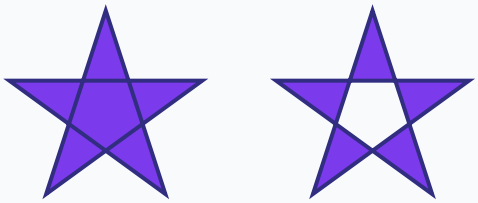


DASH PATTERNS

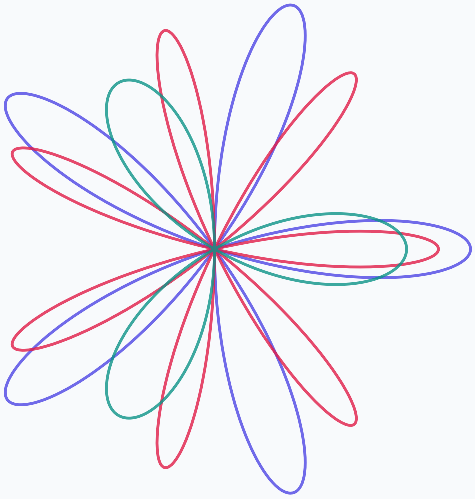


```
[] 0 d
[6 3] 0 d
[1 4] 0 d
[12 3 2 3] 0 d
[8 4] 6 d
[0.1 5] 0 d
```

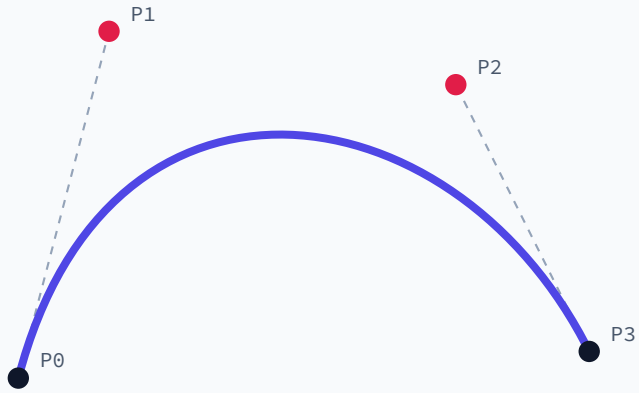
FILL RULES



PARAMETRIC ROSES (CUBIC BÉZIERS)

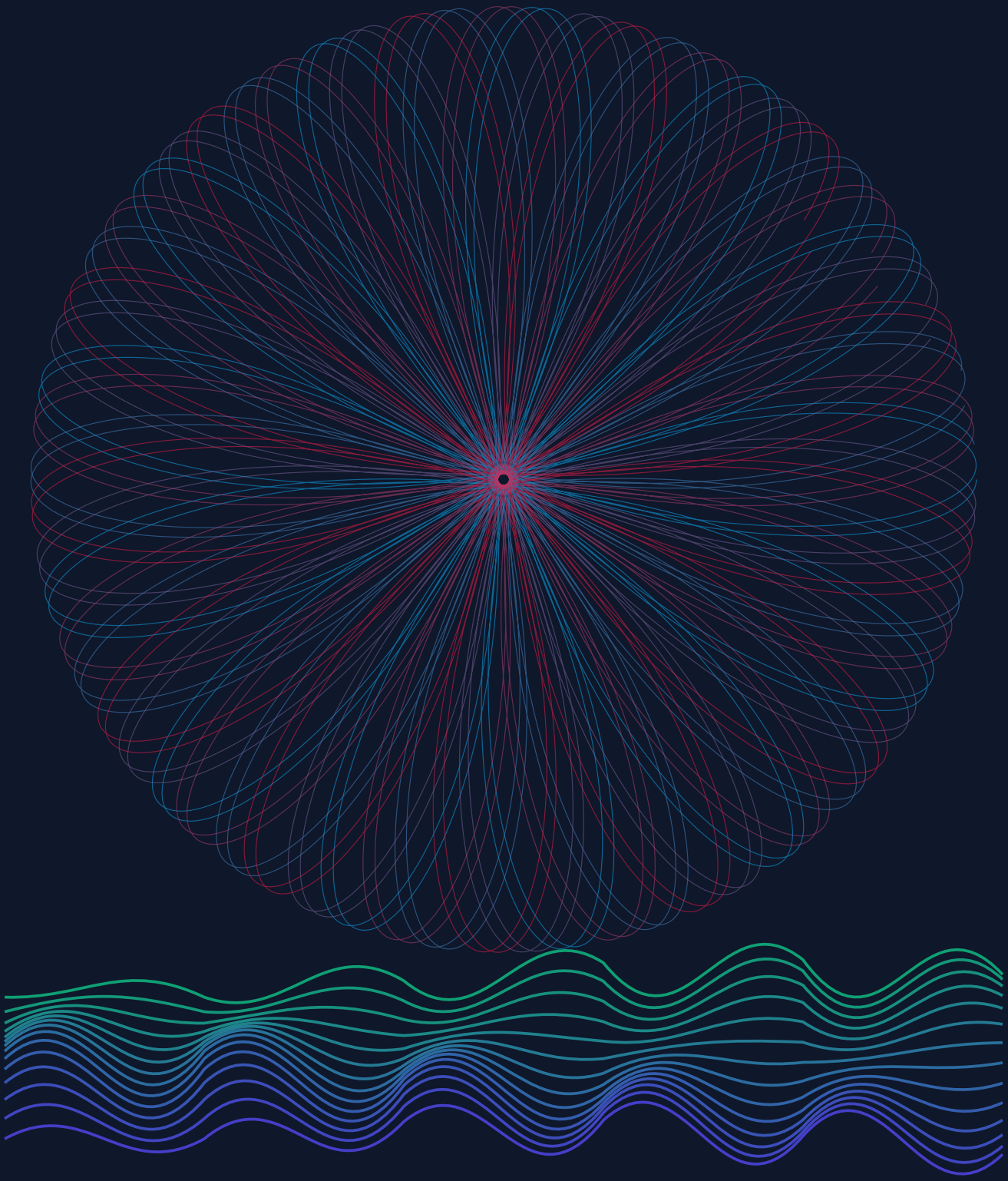


ANATOMY OF A CUBIC BÉZIER



$$B(t) = (1-t)^3P0 + 3(1-t)^2t\,P1 + 3(1-t)t^2\,P2 + t^3P3$$

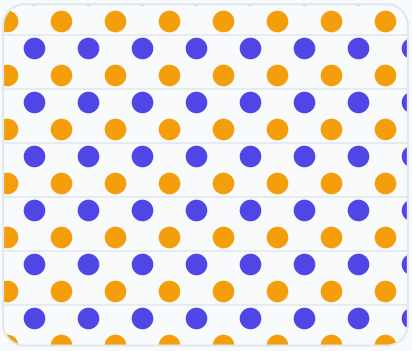
Bézier art



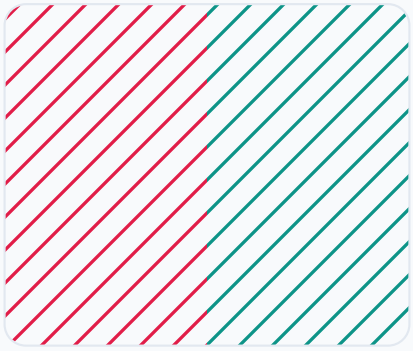
Spirograph (hypotrochoid, 5 phase-shifted strokes with 55% stroke alpha) and 14 stitched cubic wave strands.

Patterns and shadings

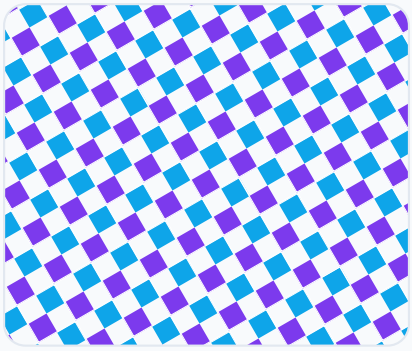
Tiling patterns (coloured and uncoloured) and all seven shading types, including Coons and tensor-product meshes.



Tiling, PaintType 1
coloured cell with its own colours



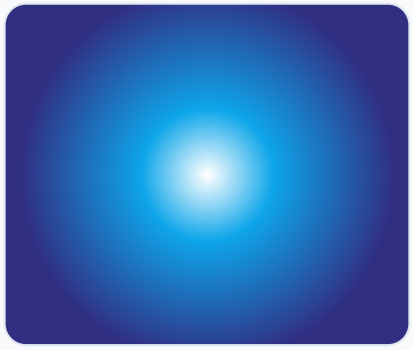
Tiling, PaintType 2
uncoloured cell; colour supplied at fill time



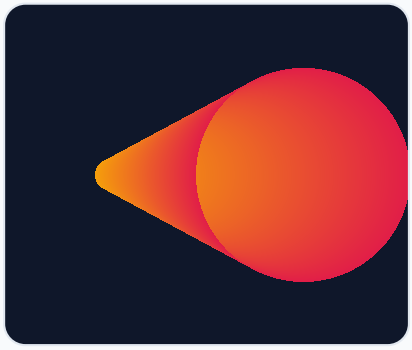
Tiling with /Matrix
pattern space rotated by 30 degrees



Axial (type 2)
three-stop stitching function



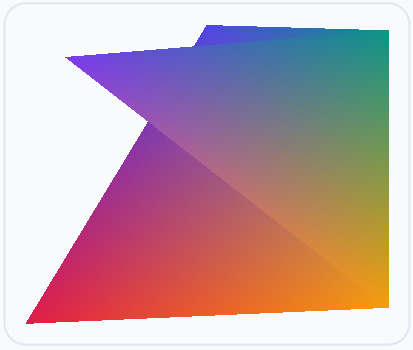
Radial (type 3)
concentric circles, extended



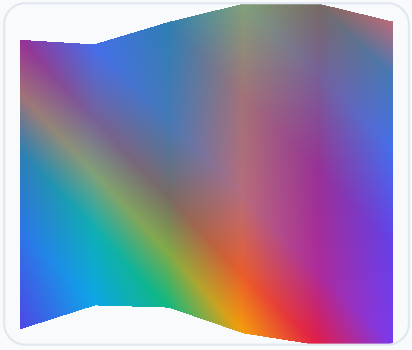
Radial, cone
two circles of different radii, no extension



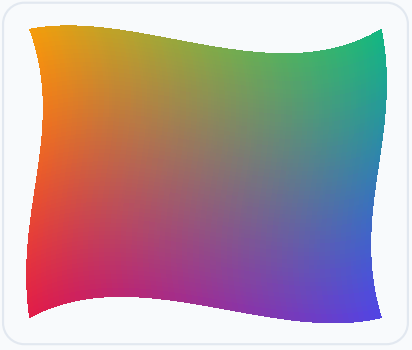
Function-based (type 1)
PostScript calculator function f(x, y)



Free-form triangle mesh (type 4)
Gouraud-shaded triangles with edge flags



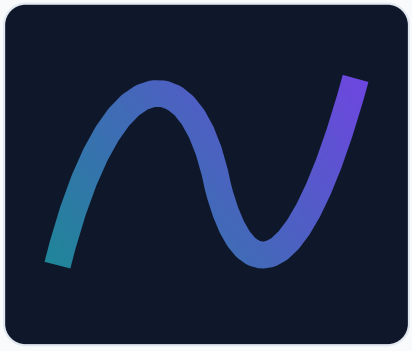
Lattice-form mesh (type 5)
grid of vertices with bilinear colour



Coons patch mesh (type 6)
one patch, four curved boundary edges



Tensor-product patch (type 7)
sixteen control points incl. interior

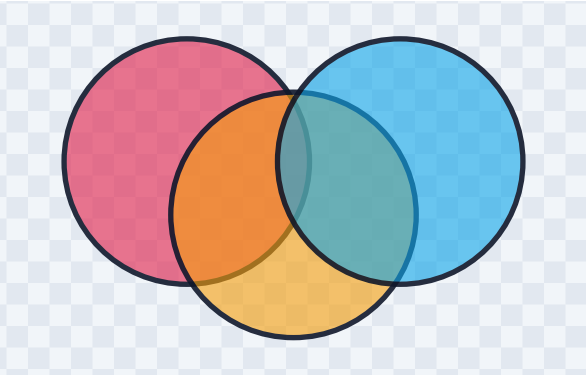


Shading pattern (PatternType 2)
a shading used as the stroke colour

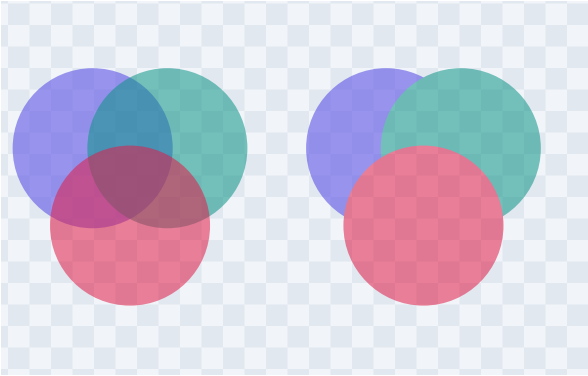
Transparency

PDF 1.4 transparency model: alpha, blend modes, soft masks and groups, composited over a checkerboard.

CONSTANT ALPHA (CA / CA)



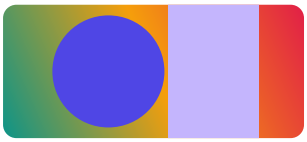
GROUP COMPOSITED AS ONE OBJECT



per-object ca 0.55

isolated knockout group, ca 0.55

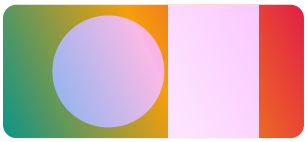
BLEND MODES (BM)



Normal



Multiply



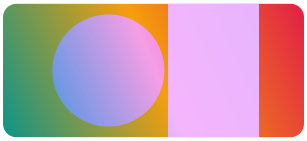
Screen



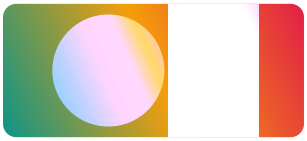
Overlay



Darken



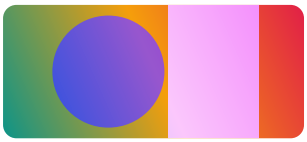
Lighten



ColorDodge



ColorBurn



HardLight



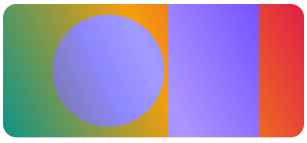
SoftLight



Difference



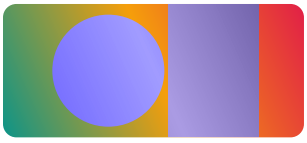
Exclusion



Hue



Saturation



Color



Luminosity

SOFT MASKS (SMASK)



/S /Luminosity



/S /Alpha

Colour spaces

Every row paints through a different PDF colour space; the reader converts to its output space.

DeviceGray
0 g and 1 g ramp



DeviceRGB
additive; sRGB-like



DeviceCMYK
C, M, Y, K ramps



CalGray
gamma 2.2, D65 white



CalRGB
colorimetric RGB



Lab
L* 55, a* from -90 to +90



ICCBased
embedded sRGB profile, N = 3



Indexed
8-entry palette



Separation
spot colour BrandIndigo, alt. CMYK



Separation
spot colour Gold, alt. RGB



DeviceN
two colorants, calculator tint transform



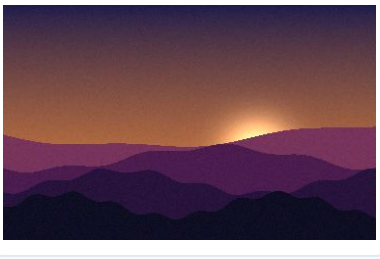
SPOT COLOUR TINTS (SCN ON A SEPARATION SPACE)



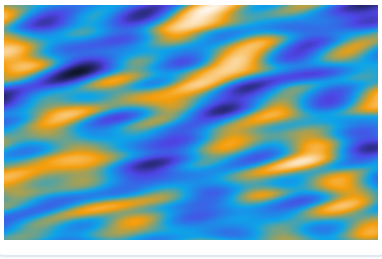
100% BrandIndigo | 75% BrandIndigo | 50% BrandIndigo | 25% BrandIndigo | 10% BrandIndigo

Image gallery

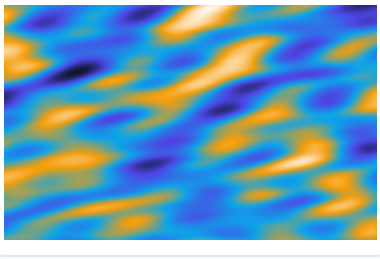
Twelve image XObjects (plus one inline image) exercising filters, colour spaces and masking.



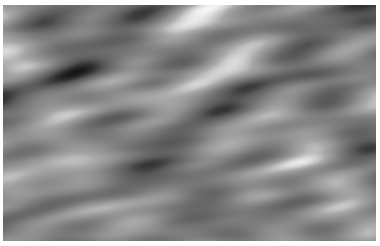
DCTDecode
baseline JPEG, passed through



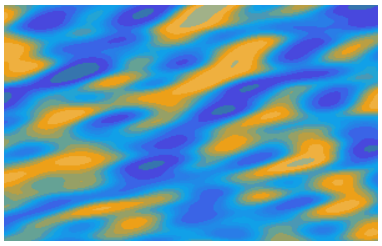
FlateDecode RGB
8-bit RGB, zlib



Flate + PNG predictor
/Predictor 15 (Up filter)



DeviceGray
8-bit, 1 component



Indexed, 4 bpc
16-colour palette, packed nibbles



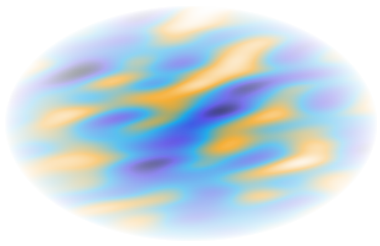
Stencil ImageMask
1-bit, painted in the fill colour



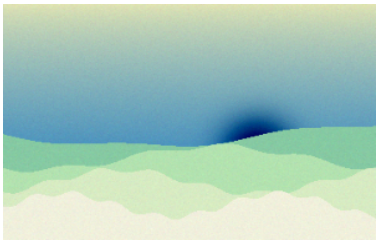
Explicit /Mask
1-bit stencil clips a photograph



Colour-key mask
/Mask [0 0 255 255 0 0] removes pure green



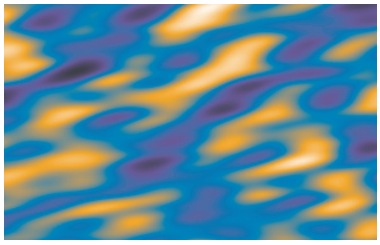
SMask
8-bit soft alpha channel



Decode [1 0 1 0 1 0]
inverts every component

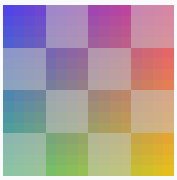


Interpolate true
16 x 16 px, smoothed on upscale



CMYK JPEG
DeviceCMYK + /Decode for Adobe inversion

Inline image (BI / ID / EI)



nearest-neighbour



/I true

Annotations

Every annotation carries an appearance stream, so it looks identical in any viewer.

Portable documents keep their meaning when content is reviewed. Reviewers mark passages with highlights, underlines and strike-outs, attach sticky notes, draw freehand ink, and stamp approvals without touching the page content beneath. Each markup is a ~~separate object~~ that can be listed, edited or removed.

Text annotation (sticky note)

URI link
Opens https://github.com in a browser

GoTo link
Jumps to page 2 at an explicit /XYZ

Free text: typeset in the annotation's own font.

APPROVED



Ink annotation: freehand stroke stored as an /InkList polyline.



Square **Circle** **Line** **Polygon**

Link target

You followed the GoTo link on page 1. The destination is an explicit /XYZ array, [page /XYZ left top zoom], with zoom null so the viewer keeps its current magnification.

Event registration

A fillable PDF form. Every widget has normal (and where relevant down) appearance streams.

FULL NAME

Ada Lovelace

EMAIL

ada@example.org

ORGANISATION

Analytical Engines Ltd

POSTAL CODE (COMB, 8 CELLS)

S W 1 A 1 A A

COMMENTS (MULTILINE)

Looking forward to the workshop on document engineering. Please seat me near the demo stage and send the slides afterwards.

TICKET TYPE (COMBO BOX)

Speaker

TRACK (LIST BOX)

Rendering
Parsing
Editing

OPTIONS (CHECK BOXES)

- [x] Subscribe to the newsletter
- [ ] Vegetarian lunch
- [x] Print my badge in advance

DAY (RADIO BUTTONS)

- ( ) Thursday
- (•) Friday
- ( ) Saturday

Submit | Reset form

Submit posts FDF; Reset clears values.

Contents

Open the bookmarks panel to see the outline. Each entry below is a GoTo link to a named destination; the printed page numbers are page labels (roman, decimal, then an A- prefixed appendix).

Foreword ii

Why navigation matters in long documents.

Introduction 1

Bookmarks, destinations and labels: three views of one idea.

Objects and syntax 2

Eight object types, indirect references and the xref table.

Page content 3

Content streams hold operators; resources are looked up by name.

Fonts and text 4

Simple and composite fonts, encodings, CMaps and the text-state parameters.

Appendix: Glossary A-1

Short definitions of the terms used in this sample.

Foreword

Why navigation matters in long documents.

Navigation targets are resolved lazily: a viewer reads the name tree, finds the destination array, then scrolls to the page, honouring the fit mode in the array. Because destinations are named, pages can be reordered without breaking links elsewhere.

← Contents

Introduction

Bookmarks, destinations and labels: three views of one idea.

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← Contents

Objects and syntax

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Page content

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← Contents

Fonts and text

Simple and composite fonts, encodings, CMaps and the text-state parameters.

Navigation targets are resolved lazily: a viewer reads the name tree, finds the destination array, then scrolls to the page, honouring the fit mode in the array. Because destinations are named, pages can be reordered without breaking links elsewhere.

← Contents

Appendix: Glossary

Short definitions of the terms used in this sample.

Navigation targets are resolved lazily: a viewer reads the name tree, finds the destination array, then scrolls to the page, honouring the fit mode in the array. Because destinations are named, pages can be reordered without breaking links elsewhere.

← Contents

PAGE BOXES / SAMPLE 13

Five boxes, one page.

MediaBox `0 0 612 792`
Physical sheet, the largest extent.

CropBox `18 18 594 774`
What viewers show and print; this page opens at this size.

BleedBox `27 27 585 765`
Where artwork must extend to survive trimming.

TrimBox `36 36 576 756`
The finished page after cutting, marked in white.

ArtBox `72 72 540 720`
The meaningful content, kept clear of the edge.

/Rotate 90

The page dictionary stores the content upright; the viewer turns the whole page clockwise by this many degrees at display time. The arrow below always points to the page's top edge.

TL

TR



content top

BL

BR

/Rotate 180

The page dictionary stores the content upright; the viewer turns the whole page clockwise by this many degrees at display time. The arrow below always points to the page's top edge.

TL

TR



content top

BL

BR

/Rotate 270

The page dictionary stores the content upright; the viewer turns the whole page clockwise by this many degrees at display time. The arrow below always points to the page's top edge.

TL

TR



content top

BL

BR

Embedded files

Three files ride inside this PDF. Open the attachments panel, or click a paperclip.

readings.csv

Sensor readings, five columns

`text/csv · 592 bytes · /AFRelationship /Data`

notes.md

Authoring notes in Markdown

`text/markdown · 150 bytes · /AFRelationship /Supplement`

mark.svg

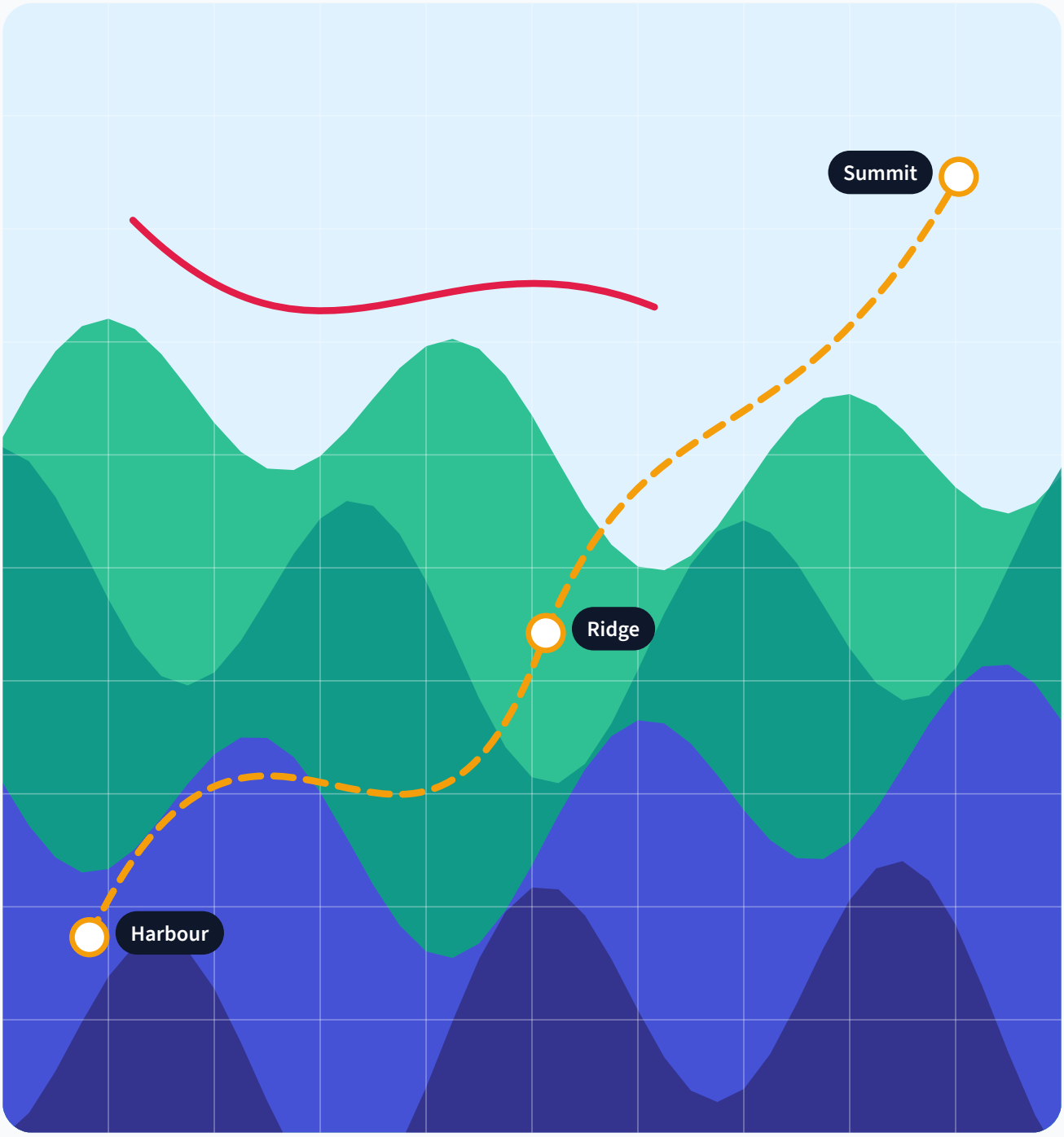
Vector logo mark

`image/svg+xml · 222 bytes · /AFRelationship /Source`

Each file specification carries /EF with a Flate-compressed stream plus /Params (Size, MD5 CheckSum, dates). /AF at the catalog level marks them as associated files, as PDF/A-3 and PDF 2.0 require.

Layers

Toggle layers in the viewer's Layers panel. The alternate palette starts hidden.



Visible when Routes + Labels on

LAYER USAGE

On: Terrain, Grid, Routes, Labels. Off: Alternate palette.
Day / Night is a radio group; the watermark prints only.

TrueType specimen

Bitstream Vera Sans embedded as glyf-outline TrueType, in both PDF flavours.

Hamburgefonts

Quick brown fox

Composite font: /Type0 + /CIDFontType2 + /CIDToGIDMap /Identity + FontFile2 subset

SIZE LADDER

The five boxing wizards jump quickly. 6 pt

The five boxing wizards jump quickly. 8 pt

The five boxing wizards jump quickly. 10 pt

The five boxing wizards jump quickly. 12 pt

The five boxing wizards jump quickly. 16 pt

The five boxing wizards jump. 20 pt

The five boxing wizards jump. 24 pt

SIMPLE FONT (/TRUETYPE, WINANSIENCODING) - CODES 32 TO 126

! " # $ % & ' ( ) * + , - . / 0 1 2 3 4 5 6 7 8 9 : ; < = > ? @

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z [\ ] ^ _ `

a b c d e f g h i j k l m n o p q r s t u v w x y z { | } ~

Simple fonts address glyphs through 8-bit codes; the viewer maps codes to glyphs via the cmap.
© 2026 ° ± × é ü •—

Bitstream Vera is a free typeface with a permissive licence: it may be embedded, subset and redistributed. TrueType outlines are quadratic splines; this page renders them as filled paths in the page's own coordinate space.

ENCRYPTION / AES-256

Board briefing

PASSWORD PROTECTED

HOW TO OPEN THIS FILE

This document is encrypted with the AES-256 security handler (V5, R6). The document-open password is user123 (owner password: owner-user123).

WHAT IT DEMONSTRATES

A /Encrypt dictionary with /CF /StdCF using /CFM /AESV3, a 32-byte file key wrapped by the user and owner passwords (/UE, /OE), and a /Perms entry. Every string and stream is encrypted except the cross-reference data.

WHY IT MATTERS

Opening this sample verifies that a reader performs the ISO 32000-2 Algorithm 2.A key derivation (the iterated SHA-256/384/512 hash) and AES-CBC decryption with a per-object IV.

ENCRYPTION / PERMISSIONS

Restricted release

OPENS FREELY · RESTRICTED

HOW TO OPEN THIS FILE

No password is needed to read it (empty user password). The owner password is owner123 ; it lifts every restriction.

RESTRICTIONS APPLIED

Printing is limited to low resolution; copying text, editing, annotating, form filling and page assembly are all disallowed. These are the /P permission bits, enforced cooperatively by the viewer.

WHY IT MATTERS

AES-256 (V5, R6) with an empty user password and a non-empty owner password is the common 'view-only' protection. A reader must derive the file key with the empty password and expose the permission flags to the application.

Writing accessible PDFs

A tagged PDF carries a logical structure beside its visual one, so assistive technology can read it in the right order.

What makes a document accessible

Marked content in the page stream is linked to structure elements through the parent tree. Reading order comes from the structure tree rather than from drawing order, and decoration such as rules and page furniture is marked as an artifact so it is skipped.

- Set the document language and title (/Lang, /Title, DisplayDocTitle).
- Mark every heading with H1 to H6 and keep the levels in order.
- Give each figure alternative text with /Alt.
- Use real table structure: Table, TR, TH with scope, TD.

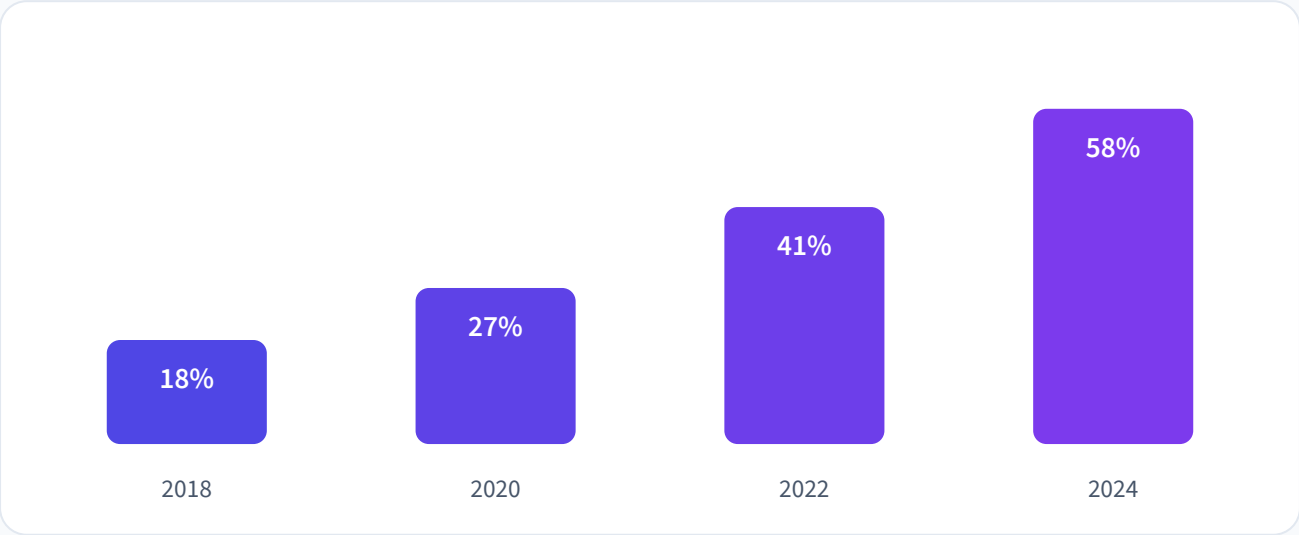


Figure 1. Illustrative share of tagged PDFs by year (sample data).

Structure types at a glance

Element	Purpose	Example
Document	Root of the tree	Whole file
H1 – H6	Headings	Chapter title
P	Paragraph	Body text
L / LI / LBody	Lists	Bullets
Figure	Images, charts	Chart with /Alt
Table / TR / TH / TD	Tabular data	This table
Artifact	Not content	Running header

Role mapping (/RoleMap) can alias custom tags onto the standard ones. Here, Callout maps to P.

NORTHLIGHT RESEARCH

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ANNUAL REPORT · 2026 EDITION

Not for redistribution — a sample document

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03 Outlook 5

04 Method and sources 6

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Executive summary

KEY MESSAGE

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Three forces explain its longevity: a small, fully specified imaging model; a conservative file structure that tolerates partial damage; and an ecosystem of independent implementations that keep each other honest. Each is examined in the sections that follow.

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> *“Longevity is a design choice, not an accident of adoption.”*
>
> — Dr. Imani Okafor, lead author

4,812	97.3 %	12	30 yrs
documents analysed	rendered identically	format clauses clarified	of backward compatibility

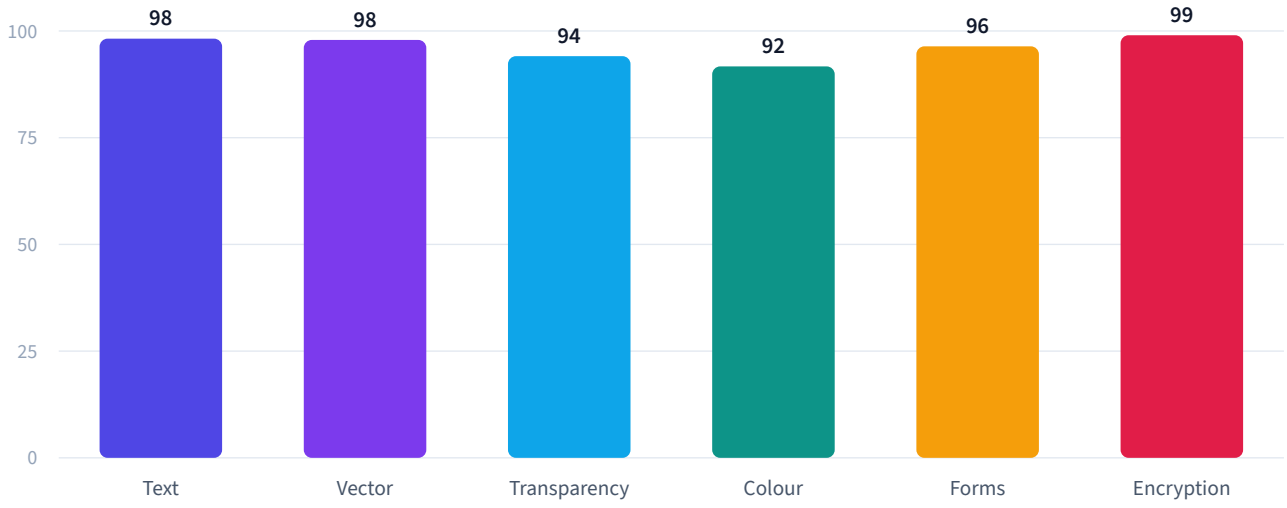
Findings

Compatibility improved across every feature family we measured. The table lists the share of corpus documents that used each feature and the share rendered without visible difference.

FEATURE FAMILY	USAGE	IDENTICAL
Text and fonts	99.1 %	98.2 %
Vector graphics	87.4 %	97.9 %
Transparency	41.6 %	94.1 %
Colour management	33.8 %	91.7 %
Forms and annotations	22.5 %	96.4 %
Encryption	9.3 %	99.0 %

FIGURE 1

Identical rendering by feature family, %

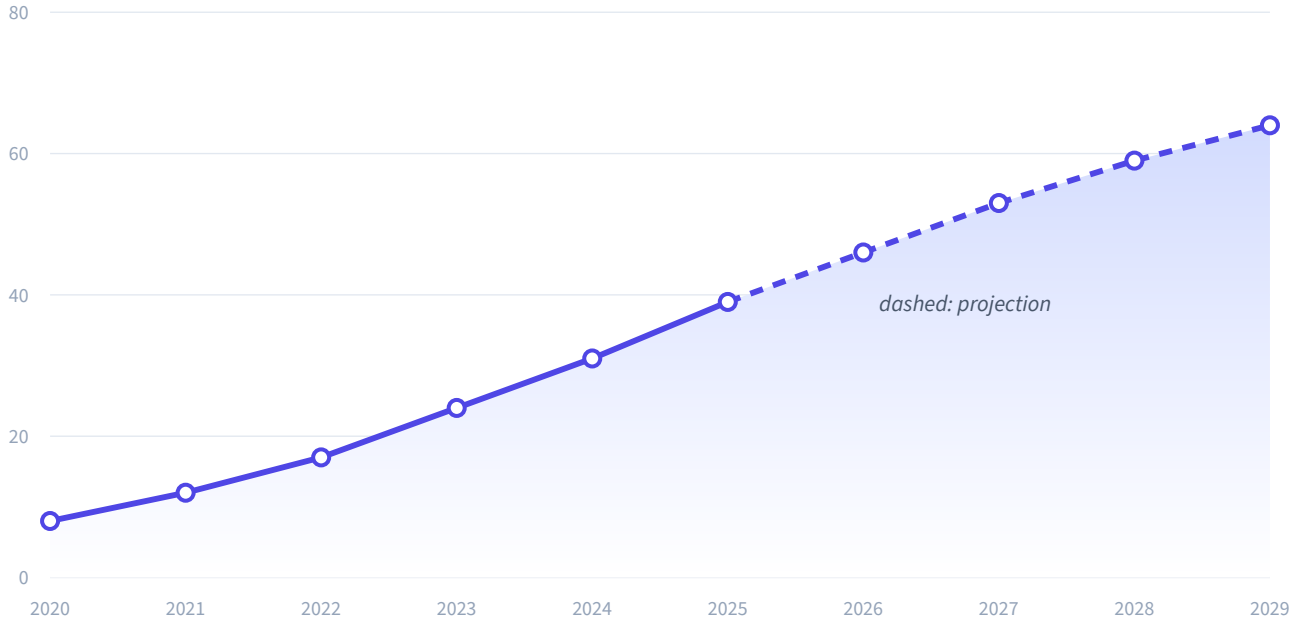


Outlook

Adoption of tagged, accessible documents is the clearest growth story. If the current trend holds, a majority of new documents will carry a structure tree by 2029.

FIGURE 2

Share of new documents with a structure tree, %



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REFERENCES

ISO 32000-2:2020	Document management — Portable document format — Part 2: PDF 2.0
ISO 14289-1:2014	PDF/UA-1 — accessible PDF
ISO 19005-4:2020	PDF/A-4 — archival
RFC 8118	The application/pdf media type

Colophon

Set in Source Serif 4 and Source Sans 3 (embedded CFF subsets). Charts are drawn with path operators and shadings; there are no raster images in this file.

GOOD TO KNOW

Allergies

Ask us about any allergen; our menu marks the common ones.

Private dining

The Lantern Room seats up to 24 for celebrations.

Sustainable

Every fish is sourced from certified day-boat fisheries.

Gift cards

Available in any amount at the counter or online.

FIND US

Visit the Quay

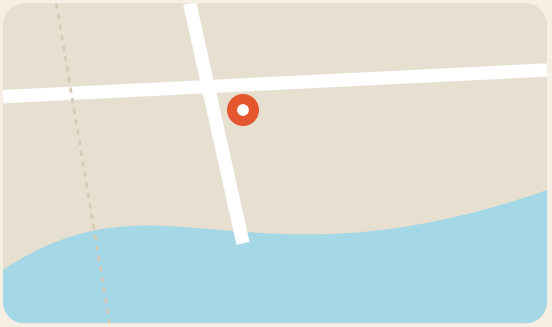
Our kitchen sits on the waterside, five minutes from the ferry landing. Reservations are recommended at weekends and during festival weeks.

ADDRESS
12 Harbour Row, Bristol BS1 4RB

OPEN
Tue–Sun, 12:00 – 22:30

PHONE
+44 117 496 0123

WEB
harbourandhearth.example



BRISTOL · EST. 1987

Harbour & Hearth

Seafood, fire & good company

01

To start

Smoked mackerel pâté 7.5
toasted sourdough, pickled cucumber

Charred octopus 11
romesco, chorizo crumb, lemon

Harbour chowder 9
clams, potato, cream, chives

Burrata & peach 8.5
basil oil, black pepper

02

From the fire

Whole sea bream 26
fennel, salsa verde, new potatoes

Wood-fired prawns 22
garlic butter, chilli, flatbread

Dry-aged ribeye 34
bone marrow butter, watercress

Hearth cauliflower 17
tahini, pomegranate, dukkah

03

To finish

Burnt honey tart 8
crème fraîche, sea salt

Chocolate pot 8
olive oil, flaked salt

Lemon posset 7.5
shortbread, thyme

Local cheeses 12
quince, oatcakes

Prices in pounds sterling, service not included. Please tell us about allergies.



Northlight Studio

Design & engineering for the web

Invoice

INVOICE NO.	ISSUED	DUE	PO REFERENCE
INV-2026-0142	15 January 2026	14 February 2026	PO-88213

FROM

Northlight Studio Ltd
14 Harbour Row
Bristol BS1 4RB
United Kingdom
VAT GB 123 4567 89

BILLED TO

Meridian Analytics Inc.
Attn: Priya Raman
500 Market Street, Suite 12
San Francisco, CA 94105
United States

DESCRIPTION	QTY	RATE	AMOUNT
Brand identity system Logo suite, colour and type guidelines	1	4,800.00	**4,800.00**
Website design 12 templates, responsive, design tokens	1	7,200.00	**7,200.00**
Front-end build Static site, accessibility audit (WCAG 2.2 AA)	64	95.00	**6,080.00**
Illustration set 24 spot illustrations, vector + PNG	24	120.00	**2,880.00**
Content strategy workshop Two half-day sessions	2	850.00	**1,700.00**
Print collateral Brochure, business cards, letterhead	1	1,950.00	**1,950.00**

PAYMENT

Bank	Northlight Studio Ltd, Monzo Business
IBAN	GB29 MONZ 0400 0412 3456 78
BIC	MONZGB2L
Reference	INV-2026-0142

Subtotal	24,610.00
VAT 20 %	4,922.00
Total due (USD)	**$29,532.00**

Payment is due within 30 days. Late payments incur 1.5 % monthly interest.
Thank you for your business; it was a pleasure working with the Meridian team.

AM

CONTACT

EMAIL
ada@marlowe.example

PHONE
+44 20 7946 0123

WEB
marlowe.example

BASED IN
Bristol, UK

SKILLS

- PDF & graphics
- C# / .NET
- Python
- Typography
- Accessibility
- Rust

LANGUAGES

English	Native
French	Fluent
German	Conversational

INTERESTS

Letterpress, ultramarathons, hand-drawn maps and the odd 1990s demoscene remake.

Ada Marlowe

SENIOR GRAPHICS ENGINEER

Engineer who turns specifications into pixels. Twelve years building document renderers, font pipelines and print tooling; happiest where PDF operators meet careful typography.

EXPERIENCE

2021 – NOW

Principal Engineer, Graphite Docs

Led the rendering core rewrite: 4× faster page rasterisation, full transparency model, CFF and TrueType font stacks. Mentored a team of six.

2017 – 2021

Staff Engineer, Quill & Co.

Designed the colour-managed PDF export path (ICC, DeviceN, soft masks) used for 40 M documents a year.

2013 – 2017

Software Engineer, Foundry Print

Built imposition and trapping tools; shipped a PostScript-to-PDF distiller and an accessibility checker.

2011 – 2013

Junior Developer, Pixelbox

Maintained a Flash-to-PDF converter, then migrated it to a cross-platform C++ library.

EDUCATION

2008 – 2011 — **BSc Computer Science (First)**, University of Bath

2007 — **A-levels: Maths, Further Maths, Art**, Redland High School

SELECTED PROJECTS

`csharp-pdf` — A from-scratch PDF parser, renderer and editor in C#.

`inkwell` — Font subsetter and CFF-to-glyf converter.

`loupe` — Pixel-diff harness for cross-renderer comparison.

PORTABLE DOCUMENT FESTIVAL · 2026

PAGES IN MOTION

every operator counts

WHEN

12 – 14 June 2026

WHERE

The Old Print Hall, Bristol

TALKS ON

Rendering · Fonts & text · Colour · Accessibility · Security · Editing

FINANCE / FY 2026

Performance dashboard

Updated 31 Dec 2026 · all figures in $ thousands

REVENUE
$786k
+18.4 %

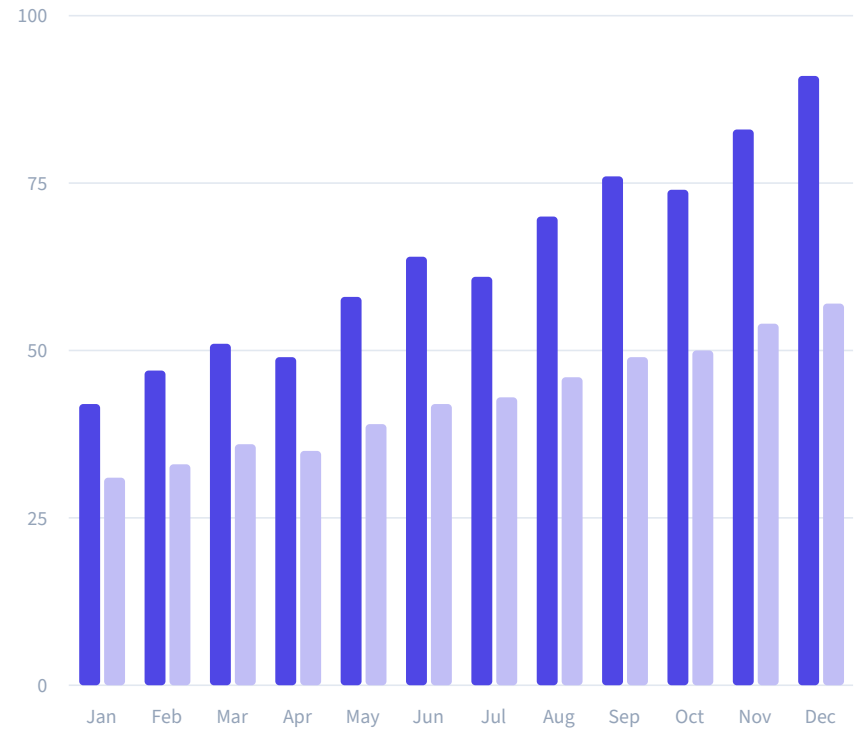
GROSS MARGIN
41.2 %
+2.1 pt

ACTIVE CUSTOMERS
4,812
+9.7 %

CHURN
1.9 %
-0.3 pt

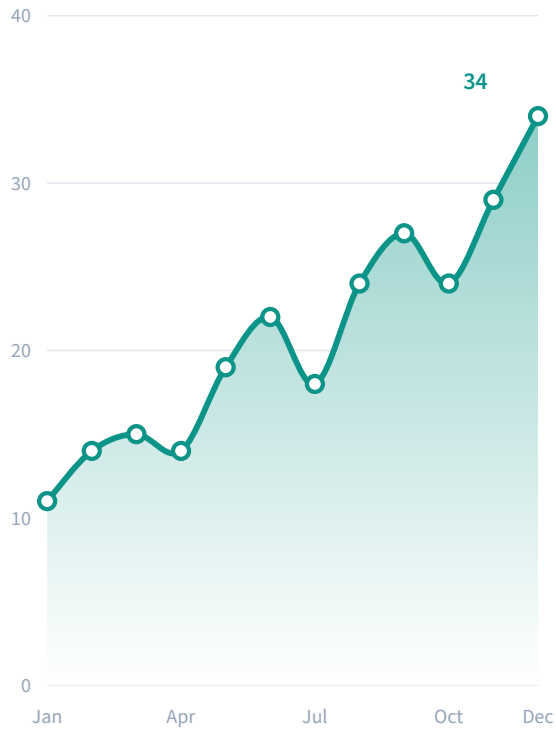
Monthly revenue vs cost

Clustered bars, $k



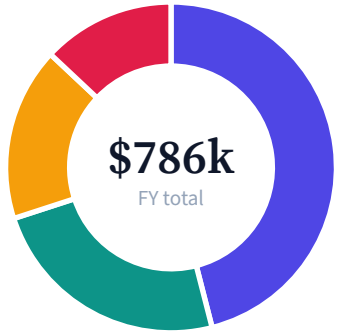
Margin trend

Revenue minus cost, $k



Revenue mix

Share by stream



Subscriptions	46 %
Services	24 %
Marketplace	17 %
Licensing	13 %

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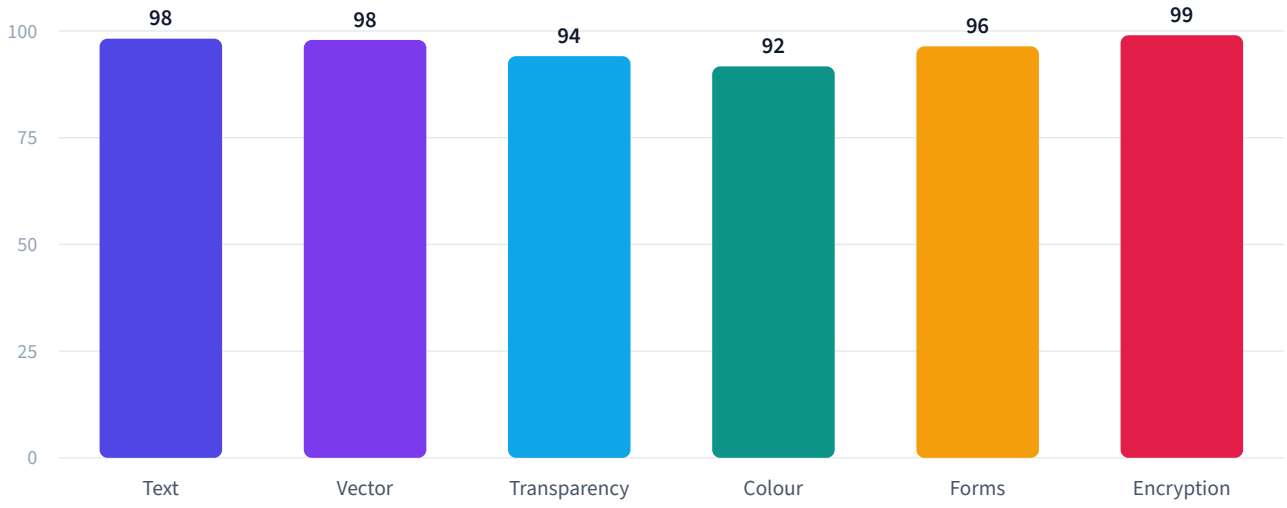
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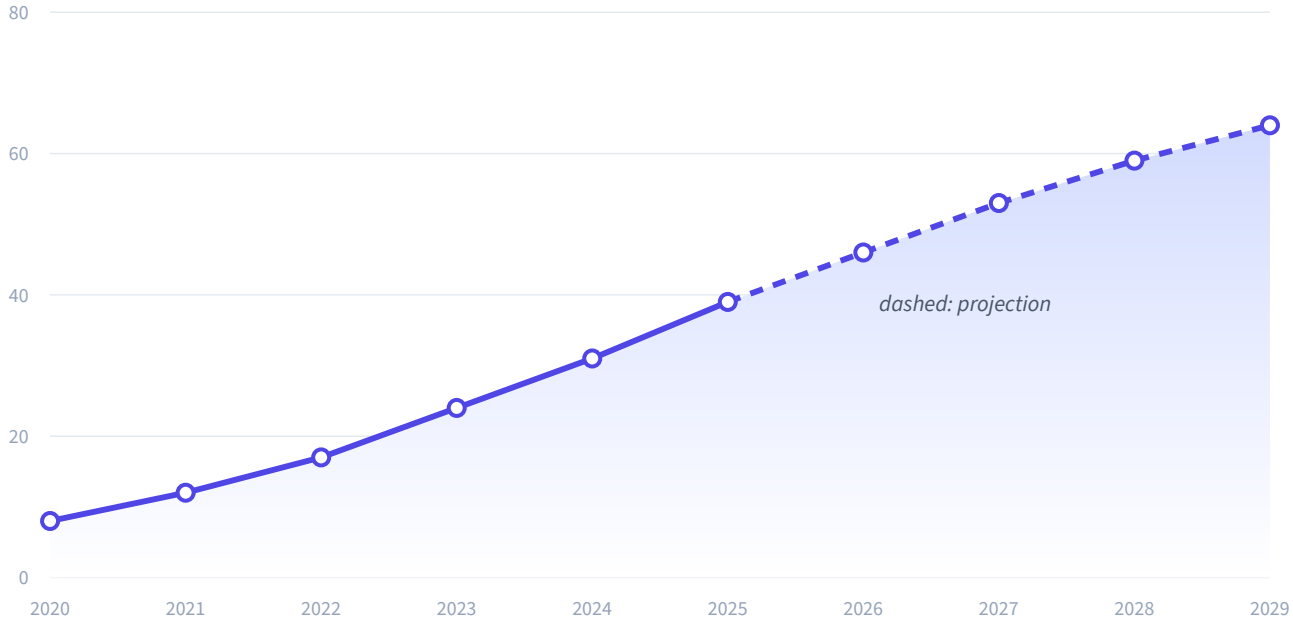


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